

=6

Research Proposal

Currently I am investigating the counting of black hole microstates using the gravitational path integral as part of my undergraduate thesis. This idea is being explored at various levels of supersymmetry. In particular, I am looking at $\mathcal{N} = 8$ and $\mathcal{N} = 4$ supergravity in four dimensions [1, 2].

Throughout these approaches, I implemented the Hardy-Rademacher-Ramanujan expansion and due to the conformal symmetry (in 2D) we see the appearance of modular forms.

These beautiful connections between number theory, gravity and string theory have been a source of fascination for me. I would like to further explore these connections in a PhD.

Throughout my spare time I have been reading a number of related texts such as Polchinski's String Theory books. This peaked my interest in Mirror Symmetry and Calabi-Yau manifolds. It is in that context where I discovered the work of Albrecht Klemm.

His work on topological string theory, Gromov–Witten invariants, special geometry, and mirror symmetry forms a natural theoretical foundation for extending my current interests into a broader geometrical and mathematical framework.

Background and Connection to Previous Work

The microscopic interpretation of black hole entropy in string theory originating from the seminal work of Strominger and Vafa which links gravitational thermodynamics with D-brane bound state counting. Modern refinements, such as supersymmetric localization and gravitational modularity, have provided increasingly precise enumerative tools.

Parallel to this development, the geometry of Calabi–Yau manifolds and mirror symmetry has yielded exact formulas for worldsheet instanton corrections, Gopakumar–Vafa invariants, and topological string partition functions. The modular properties of these generating functions echo the modular structure underlying 2D conformal field theories and supergravity microstate degeneracies.

I hope to explore the following possible research directions (subject to refinement in collaboration with Prof. Klemm):

- Modular and automorphic structures in topological string partition functions, and their relation to black hole entropy formulas.
- The interplay between Gopakumar–Vafa invariants, BPS state counts, and microscopic entropy, especially in compactifications with reduced supersymmetry.
- Mirror symmetry and enumerative invariants as tools for extracting physical microstate counting data, possibly leveraging the geometry of specific Calabi–Yau threefolds.
- Connections between quantum geometry, special geometry, and the emergence of modularity in gravitational path integrals.

Personal Preparation

Throughout my undergraduate studies (primarily my thesis), I have gained foundational knowledge in the following areas which I believe will be useful in pursuing this research:

- Two dimensional conformal field theory
- Supersymmetric black holes
- Gravitational partition functions and saddle point methods
- Modular and automorphic forms
- Number theoretic expansions such as the Hardy-Radmeacher-Ramanunjan formula
- The basics of string theory

In addition to my formal coursework, I have studied Polchinski's String Theory volumes I and II, as well as David Tong's lecture notes on string theory. his independent learning led me to Prof. Klemm's work, particularly regarding special geometry, enumerative invariants, and modularity in topological strings.

I have also developed pertinent mathematical skills, including Lie algebras, representation theory, differential geometry, and some exposure to complex geometry which are foundational tools I intend to strengthen during doctoral research.

Conclusion

The profound connections between number theory, geometry, and fundamental aspects of quantum gravity are intellectually compelling and, I believe, still far from fully understood. I am eager to work within the research environment that Prof. Klemm and collaborators provide in order to advance these ideas and grow as a mathematical physicist. I am highly motivated to contribute to this area of research within a PhD programme at your Sheffield university.

Bibliography

- [1] Luca V. Iliesiu, Sameer Murthy, and Gustavo J. Turiaci. Black hole microstate counting from the gravitational path integral, 2022.
- [2] Gabriel Lopes Cardoso, Abhiram Kidambi, Suresh Nampuri, Valentin Reys, and Martí Rosselló. The gravitational path integral for $n=4$ bps black holes from black hole microstate counting. *Annales Henri Poincaré*, 24(10):3305–3346, March 2023.